

Cost-to-Serve Curve

SLIBRARY 07 · SHEET 1 / 5

Unit cost falling with automation

PROFESSIONAL DEFINITION

The Cost-to-Serve Curve shows how the unit cost of serving a customer or transaction falls as automation and AI absorb increasing volume, revealing scale economics.

WHO DEVELOPED IT

Rooted in cost-accounting and operations practice, with cost-to-serve analysis popularised in supply-chain and service-economics literature.

WHY IT WAS DEVELOPED

It was developed to quantify how automation changes the economics of service delivery, distinguishing fixed investment from falling marginal cost.

FIGURE 1 · COST-TO-SERVE CURVE

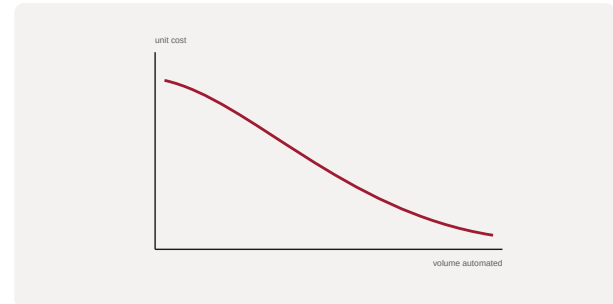


Figure 1. A schematic of the Cost-to-Serve Curve as applied in BARBM312 (illustrative).

HOW IT IS USED

Unit cost is plotted against automated volume; the curve's shape informs pricing, capacity and the payback on automation investment.

WHEN IT IS USED

When evaluating automation ROI and designing scalable service models.

BY WHOM IT IS USED

Operations, finance and pricing leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes the scale economics of AI-enabled service explicit, linking automation investment to declining unit cost.

RELATED FRAMEWORKS IN THIS COURSE

Process Cycle Efficiency · Slibrary 7

Lean Waste Lens (AI) · Slibrary 7

Throughput Accounting · Slibrary 7

Effort-Value Reallocation · Slibrary 7

SOURCES (HARVARD REFERENCING STYLE)

Cokins, G. (2001) Activity-Based Cost Management. Hoboken: Wiley.

Anderson, S.W. (2007) 'Managing costs and cost structure', Handbook of Management Accounting Research, 2.